

Multi-scale multi-instance contrastive learning for whole slide image classification

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ABSTRACT

Multi-instance learning (MIL) has become the mainstream solution for processing super-high resolution whole slide images (WSIs) with the pyramidal structure in digital pathology. Current MIL-based methods usually learn features from WSI at a specific magnification, ignoring the multi-scale information contained in the WSI and the comparative learning of global features. In addition, the lack of instance labeling can lead to weak model supervision, which may compromise the model's ability to discriminate fine-grained features, ultimately affecting bag-level feature learning. Therefore, we propose a novel multi-scale multi-instance contrastive learning framework to learn more discriminative feature representation across scales for pathological WSI classification. The proposed method begins with a two-stream feature aggregator module, which extracts both bag embeddings and selects the representative instances simultaneously. Following the bag embedding branch, a multi-scale contrastive learning module is designed to learn the global feature comparisons of WSIs across multiple scales by leveraging its inherent pyramid structure. Additionally, based on the instances selection branch, a patch-level classifier is combined with the bag-level classifier to jointly optimize the model training process, enhancing the supervision of the model. The proposed framework is evaluated on three publicly available WSI datasets, achieving an area under the curve of 95.8%, 95.5%, and 88.2%, respectively, consistently outperforming all the compared methods including single- and multi-scale ones.

1. Introduction

Histopathological diagnosis is the gold standard for diagnosing almost all types of cancers (Gurcan et al., 2009). With the current advances in scanner technology, whole-slide imaging has been widely used for the preservation of histopathology slides, i.e., scanning and digitizing entire histological slides (Khened et al., 2021). A whole slide image (WSI) is stored in the form of a pyramid containing multiple magnifications. With the rapid development of deep learning methods, computer-aided WSI analysis has become increasingly important and attracted considerable attention in the medical imaging field, largely benefiting automated diagnosis, prognosis, and therapeutic-response prediction (Xing et al., 2017; Jiang et al., 2024). However, one typical WSI has a size of $40\,000 \times 40\,000$, which makes it infeasible to compute directly using conventional deep-learning classification networks under current hardware. To address this challenge, a large number of patch-based approaches have appeared, whereby a WSI is divided into multiple patches that can be fed into a neural network for computation (Sharma et al., 2017; Coudray et al., 2018; Cheplygina et al., 2019; Campanella et al., 2019; Sun et al., 2020; Li et al., 2021; Lu et al., 2021b,a; Shao et al., 2021; Zhang et al., 2022b; Wibawa et al., 2022;

Su et al., 2022, 2023; Hu et al., 2023; He et al., 2023; Yang et al., 2024). Among them, one of the challenges with the supervised learning-based approaches (Sharma et al., 2017; Coudray et al., 2018; Mathew et al., 2023) is the need for detailed annotations of the target of interest. This is a labor-intensive and time-consuming task given that annotations of disease extent typically need to be provided by pathologists, who have a minimal amount of time to begin with. Therefore, recent studies (Cheplygina et al., 2019; Campanella et al., 2019) have started to focus on using weakly supervised learning for WSI analysis. In contrast to supervised learning, weakly supervised learning algorithms that have a good balance between cost and accuracy offer promising approaches.

In particular, the majority of weakly supervised classification methods on WSIs (Li et al., 2021; Lu et al., 2021b,a; Shao et al., 2021; Zhang et al., 2022b; Wibawa et al., 2022; Su et al., 2022) follow a multi-instance learning framework (Dietterich et al., 1997). Multi-instance learning (MIL) is one of the subclasses of weakly supervised learning, which is commonly used to solve binary classification problems with sets, where the set is called a bag and the elements of the set are called instances. Within the context of MIL, only bags have available labels, while the labels of the instances in the bag are unknown. In

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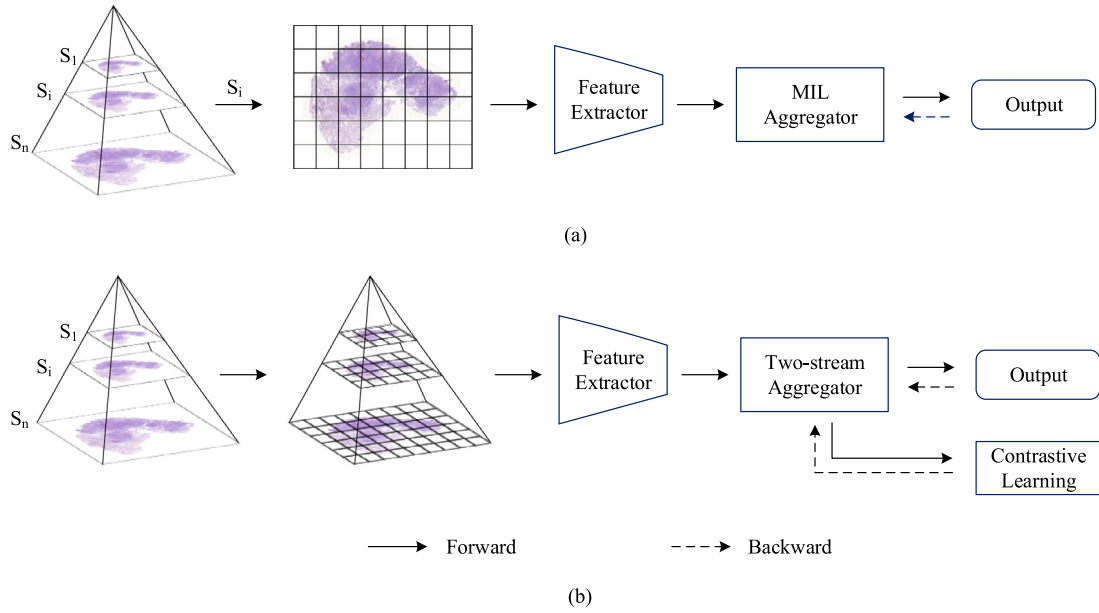


Fig. 1. (a) Common MIL model for WSI classification. Typically WSIs are stored in the form of a pyramid containing images at multiple magnifications. Common MIL frameworks typically select an image at a specific magnification level to be used in model training. The chosen image is then divided into thousands of patches, which are subsequently passed through a feature extractor to obtain instance feature vectors, followed by a MIL aggregator for aggregation and prediction. (b) Our proposed method. Compared with the traditional MIL-based methods, our proposed method utilizes WSI images at different magnifications and obtains distinguishable features of WSIs through contrastive learning.

the MIL framework for the analysis of WSI, one WSI is considered as a bag, and the patches divided from the WSI are considered as instances in the bag. If a WSI (bag) is lesion-contained (positive), then at least one patch (instance) from this WSI (bag) contains a lesion (positive). Consequently, only slide-level labels are required for WSI classification using MIL, which significantly reduces the workload of pathologists.

As illustrated in Fig. 1(a), most of the current MIL-based methods (Ilse et al., 2018; Campanella et al., 2019; Lu et al., 2021b,a; Shao et al., 2021; Zhang et al., 2022b; Wibawa et al., 2022; Su et al., 2022; Zhu et al., 2022; Wang et al., 2023; Xiong et al., 2023a; Yang et al., 2024) for the WSI analysis follow a two-stage paradigm, comprising patch-level feature extraction (i.e., the construction of a MIL problem) and the subsequent slide-level aggregation phase. Specifically, in the patch-level feature extraction stage, an image at the specific magnification of WSI is selected and divided into thousands of patches, which are fed into a feature extractor to obtain instance feature vectors. At this point, the WSI classification problem with only slide-level labels is transformed into a classification problem about a set consisting of multiple unlabeled instances, i.e., the MIL problem. In the slide-level aggregation phase, the MIL aggregator aggregates the features of all instances in the bag to obtain the slide-level embedding representation and classification results. While these approaches have yielded good results, there are still several challenges in developing suitable models for accurate WSI analysis.

Firstly, in terms of the overall framework, most previous explorations (Ilse et al., 2018; Campanella et al., 2019; Lu et al., 2021b,a; Shao et al., 2021; Zhang et al., 2022b; Su et al., 2022) have been limited to images at a single magnification, which simplifies the problem but ignores the hierarchical information contained in the WSI and the comparison of global features. Layers of WSI at different magnifications provide diagnostic information from different views ranging from tissues to cells. Analysis based on a single magnification only captures a portion of the information, so it is important to leverage the hierarchical nature of WSI to enhance the performance of the model. Some recent studies (Hashimoto et al., 2020; Li et al., 2021; Deng et al., 2022; Zhang et al., 2022a; Zhou and Lu, 2022; Liu et al., 2023; Sun et al., 2023) have considered the information conveyed by images at multiple magnifications, demonstrating that considering multi-scale information has a positive impact on analyzing WSIs. However, the majority of these

multi-scale algorithms integrate information acquired from multiple scales through simple operations such as concatenation (Zhou and Lu, 2022; Liu et al., 2023) and attention mechanisms (Hashimoto et al., 2020; Li et al., 2021; Deng et al., 2022; Sun et al., 2023), ignoring the semantic gap between different scales and global feature comparisons of WSIs, i.e., comparisons between bags.

Secondly, the absence of direct supervision for each instance presents a challenge for the MIL approach in constraining the learning process during the slide-level aggregation phase. It may result in the generated slide-level embedding representations of the WSIs that may contain a large amount of redundant information. Most of the previous studies (Ilse et al., 2018; Hashimoto et al., 2020; Shao et al., 2021; Su et al., 2022; Zhu et al., 2022) have used certain fusion strategies (e.g., attention mechanism and pooling operations) to aggregate the instance feature vectors into a slide-level representation for prediction. This is a relatively simple and straightforward process, however, if high weights are assigned to cases in irrelevant patches, this inaccurate information will be passed on to the final slide-level representation. It may reduce the discrimination of slide-level representations and result in incorrect model predictions. That is, the bag-level representation without any constraint may contain inaccurate information, which can negatively affect the accuracy and generalization of the model. Therefore, enhancing the supervisory nature of the model through reasonable strategies can further improve the performance of the MIL method. Articles (Lu et al., 2021b; Tu et al., 2022; Wang et al., 2022a; Zhang et al., 2022a) introduce a patch-level classifier that predicts instance labels while obtaining bag-level predictions to enhance the supervisibility of the model. However, some methods (Tu et al., 2022; Zhang et al., 2022a) default to slide-level labels inherited by all instances in the bag, inevitably introducing noise. In contrast, others (Lu et al., 2021b; Wang et al., 2022a) filter instances and assign pseudo-labels to them using only attention scores, which loses the diversity of instances despite reducing noise introduction.

To address these aforementioned challenges, a novel multi-scale multi-instance contrastive learning model (named MSCL-MIL) is proposed to handle the problem of classifying WSIs with only slide-level labels. The objective is to achieve discriminative feature learning for WSIs and global feature representations contrast between WSIs. As shown in Fig. 1(b), to achieve this goal, we design a two-stream feature

aggregator and introduce a contrastive learning strategy. The two-stream feature aggregator is responsible for completing the selection of representative instances while obtaining the slide-level embedding representation (see Fig. 2(b)). Following the bag embedding branch, we hire contrastive learning to participate in the training process of the model (see Fig. 2(c)), aiming to efficiently learn global features among WSIs. Specifically, the pyramid structure of the WSIs is employed to generate the positive and negative sample pairs for contrastive learning, aiming to facilitate the acquisition of common features of similar WSIs and the discriminative features to different classes of the WSIs. In addition, in order to learn the appropriate slide-level embeddings in the training processing, a patch-level classifier is introduced based on the selected instances to promote the learning of the attention weights of the patches. Notably, previous approaches (Lu et al., 2021b; Wang et al., 2022a) only utilize attention scores to select the instance, which may result in the selected instances being unitary and not representative of the distribution of the entire WSI. Conversely, we introduce a clustering algorithm to evenly select the instance in the whole WSI, which ensures that the distribution of the selected instances is consistent with the intra-bag instances, thus exposing the model to different discriminative patches from the WSI. Finally, the self-supervised learning layer and the patch-level classifier in the model is frozen to achieve the final prediction in the testing phase. The main contributions of this paper are summarized as follows:

- We propose a novel multi-scale multi-instance contrastive learning method for improved WSI classification, which uses contrastive learning to extract global features and incorporates a patch-level classifier to enhance model supervision.
- We design a two-stream feature aggregator that integrates both slide-level representations and instance-level selection. An attention mechanism generates slide-level embeddings, while a clustering algorithm is introduced to select representative instances to refine attention weights and constrain model training.
- We propose a multi-scale contrastive learning module that regards images from multiple magnifications of the same slide as positive pairs and images from different slides as negative pairs, efficiently learning discriminative features via multi-scale contrastive loss.

The remainder of this paper is structured as follows. In Section 2, we review some work related to our research. In Section 3, we introduce our proposed framework. In Sections 4 and 5, experimental details and results for evaluating the proposed method on different datasets are given. Finally, discussion and conclusions are drawn in Sections 6 and 7.

2. Related work

In this section, a brief review of closely related work in the literature is presented, including MIL methods for WSI classification and MIL methods based on contrastive learning.

2.1. MIL for WSI classification

Multi-instance learning has now been successfully applied to the analysis of WSIs, such as classification (Campanella et al., 2019; Lu et al., 2021b; Su et al., 2022), segmentation (Feng et al., 2021; Li et al., 2023) and survival prediction (Tang et al., 2024). In this case, one WSI is usually divided into numerous patches and is considered to be a bag consisting of these patches. In a typical WSI classification pipeline based on MIL, patch-level features are extracted, aggregated, and classified to generate slide-level prediction.

As a WSI comprises a vast number of patches, several methods reduce the computational burden by filtering the instances. The first strategy is to make an initial prediction of patches and filter them according to the predicted probability. For example, Campanella et al. (2019) select Top- k patches as representative instances based on the

probability that a patch is positive given by a patch-level classifier. Su et al. (2022) classify the patches into different categories based on the patch-level classifier predictions, sort them in each category according to the predicted probability values, and select the Top- k patches that were sorted top among all the instances. However, the labels of the patches are usually inherited from the labels of the slides in the above methods when using patch-level classifiers to predict the probability of patches, which inevitably introduces label noise. To solve this problem, attention-based and clustering-based instance selection methods have been proposed. Zhang et al. (2022b) filter patches based on the attention scores based on attention scores obtained from the attention mechanism in ABMIL (Ilse et al., 2018). Shi et al. (2023) filter redundant instances using a multi-head self-attention mechanism. Nevertheless, the patches selected by these attention-based methods may be monolithic, i.e., they may be from the same lesion, which does not preserve the diversity of instances. Thus, Sharma et al. (2021) and Wang et al. (2022b) utilize the K-means method to select patches, which divide patches into several clusters and then sample instances to participate in the final classification prediction. Furthermore, Lu et al. (2021b) design a patch-level clustering task to assist model training based on instance selection, which first filters the patches using attention scores and then clusters them via smooth SVM loss. However, the order of operation does not ensure the diversity of sampled instances like clustering algorithms such as K-means.

In addition, the use of bags containing features at different magnifications proved to be beneficial for the WSI classification based on MIL. Hashimoto et al. (2020) simply fuse features from patches at different magnifications of the WSI in a single bag to generate slide-level prediction through the attention mechanism and show good results on the malignant lymphoma dataset. Li et al. (2021) integrate patch features from different magnifications through a pyramid cascade strategy and perform experiments on a breast cancer dataset. Deng et al. (2022) first perform a cluster analysis of patches at low magnification of the WSI and then use features of patches at multiple magnifications in each cluster through a cross-scale attention mechanism. Diao et al. (2023) use a similarity cross entropy loss function to simultaneously learn the similarity of the information among cross-magnifications. Instead, we consider images at multiple magnifications as pairs of positive/negative samples, and learn the features of the bags better through multi-scale contrastive learning.

2.2. MIL based on contrastive learning

Contrastive learning is a subclass of self-supervised learning that aims to learn data-rich feature representations while not relying on data labels. The contrastive learning models learn discriminative features of the data by constructing similar (positive) samples versus dissimilar (negative) samples. Current applications (Li et al., 2021; Deng et al., 2022; Schirris et al., 2022; Wang et al., 2023) of contrastive learning-based MIL in WSI analysis are mostly focused on patch-level feature extractors. These studies propose the use of a contrastive learning model as a patch-level feature extractor to obtain discriminative feature representations of patches as pre-trained convolutional neural networks (CNN) are considered more suitable for natural scene image feature extraction. However, due to the extremely high resolution of WSIs, with thousands of patches available in a single WSI, pre-training a contrastive learning model as a feature extractor incurs a large computational burden. Moreover, experimental results from one study suggest that using a CNN pre-trained on ImageNet as a patch-level feature extractor may outperform self-supervised learning (Tavolara et al., 2022).

Lately, many studies learn the discriminative information of bags by contrastive learning. These methods sample a subset of patch-level feature vectors of a WSI, and the embedding representations of the different subsets obtained are used as positive/negative sample pairs for contrastive learning, which in turn learns the discriminative

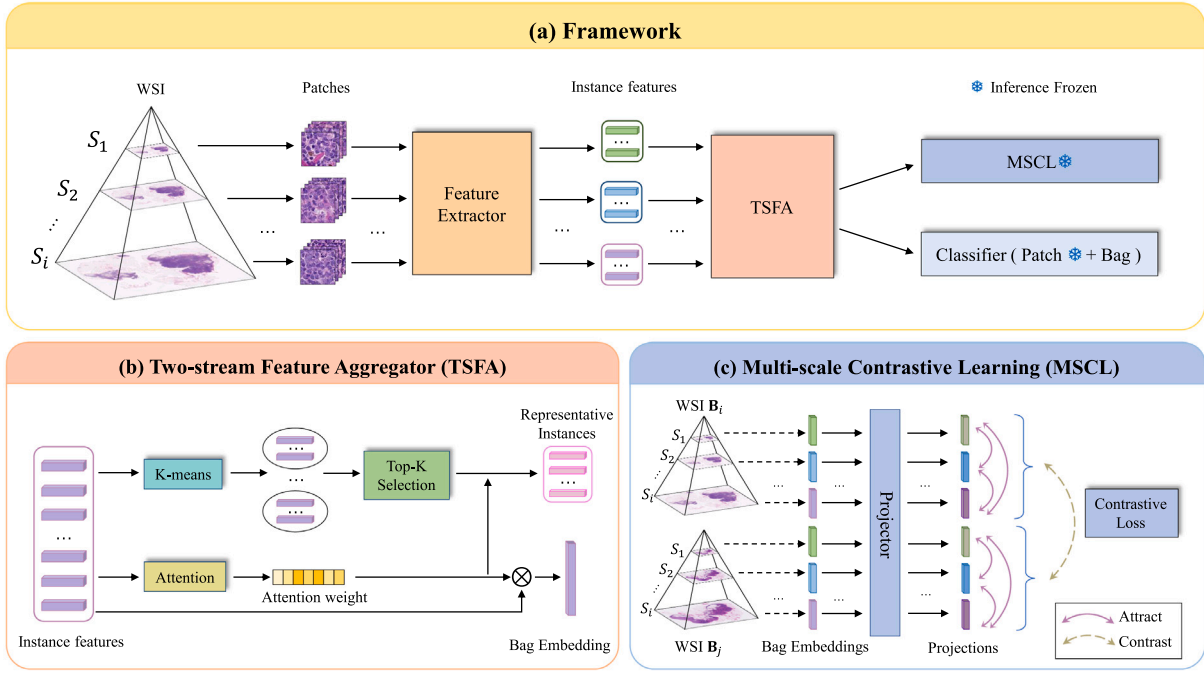


Fig. 2. Overview of the proposed MSCL-MIL. (a) The structure of our proposed method with the overall flowchart, including dividing patches, extracting instance feature vectors, and predicting slide labels. The patches (instances) at different magnifications of the WSIs are first divided and then their feature vectors are extracted by the feature extractor. The instance features are aggregated and analyzed by TSFA, and the results are fed into the MSCL module to assist in model training, and finally predicted by the classifier. (b) The details of our proposed two-stream feature aggregator (TSFA). The TSFA framework contains two branches: an instance selection branch based on clustering algorithms, which is used to select K representative instances from a large number of instances; and an instance feature aggregation branch based on the attention mechanism, which is capable of intelligently integrating the instance features to support the training and prediction of subsequent models. (c) Illustration of the proposed multi-scale contrastive learning (MSCL) module. The WSI is partitioned by patches, feature extraction and aggregation to obtain a bag-level embedding, which is fed into a multi-scale contrastive learning module. The MSCL and the patch-level classifier is frozen during the testing phase.

information of the bags (Zhu et al., 2022; Tavolara et al., 2022). In addition, Wang et al. (2022a) filter the patches in the bags according to the attention scores and thus constructs a series of new positive and negative bags, followed by bag-level contrastive learning by calculating the contrastive learning loss of the instances in the new bags. Zhang et al. (2022c) first obtain two views of each instance through data augmentation, and then constructed positive/negative pairs by concatenating the feature vectors of each view of all the instances separately to obtain the embedding representations of the bags. They subsequently learned the information contained in the bags using supervised contrastive learning loss. Their experimental results demonstrate that considering the discriminative information of bags contributes to the performance of the model.

3. Methodology

The motivation of this paper is to automatically classify WSIs using only slide-level labels. An overview of our method is shown in Fig. 2, which mainly includes the division of patches, feature extraction, two-stream feature aggregator (TSFA), and multi-scale contrastive learning (MSCL). In this section, the classification problem of WSI is first formulated as a MIL problem through patch division and feature extraction. Then the MIL model is constructed through two-stream feature aggregator and multi-scale contrastive learning. We elaborate on our components in the following subsections.

3.1. Problem formulation

In the MIL setting, the samples are structured as bags consisting of multiple instances. Let $\mathbf{B}_i = \{X_1^{(i)}, X_2^{(i)}, \dots, X_m^{(i)}\}$ represent a bag containing m instances, where $X_j^{(i)}$ denotes the j -th instance from the i -th bag. The label of the bag is represented as $Y_i \in \{+1, -1\}$, where $Y_i = +1$ means that the bag is positive, and otherwise $Y_i = -1$ means

that the bag is negative. In addition, the instances in the bag are unlabeled. The goal of a MIL model is to predict the bag label with unlabeled instances.

In this paper, one WSI is considered as a bag $\mathbf{B}_i = \{X_{1s_p}^{(i)}, \dots, X_{ms_p}^{(i)}; \dots; X_{1s_p}^{(i)}, \dots, X_{ns_p}^{(i)}\}$ and the patches $X_{js_p}^{(i)}$ from different magnifications are considered as instances, where s_p denotes the p -th magnification, and m and n are the number of patches at different magnifications, respectively. Only the label for the WSI is known, such as normal or tumor, and none of the labels for patches are known. This paper would like to predict the labels of WSIs directly by learning the information expressed by unlabeled patches. To ease the training, the pre-trained feature extractor $F(\cdot)$ is used to extract the feature $f_{js_p}^{(i)} = F(X_{js_p}^{(i)})$ of the patch $X_{js_p}^{(i)}$.

3.2. Two-stream feature aggregator

Following the conversion of a WSI classification problem into a MIL problem, a significant challenge arises in the design of an effective aggregator that is able to map instances to an embedding space to obtain a suitable bag-level representation.

Commonly used feature aggregation methods are max-pooling, average pooling, and attention mechanisms. Among them, the MIL models give strong assumptions about the structure of the bag when using max-pooling and average pooling. Average pooling considers all instances to be indistinguishable (which can be thought of as giving equal weight to all instances), while max-pooling focuses on only one instance and ignores the others. Both them are predefined and untrained. In addition, the information loss in both methods is extreme and the distributive properties in the bag are completely eliminated in the process (Chi et al., 2022). Whereas the attention mechanism continuously adjusts the importance of each instance to the bag label during training by learnable parameters, it has now become the main feature aggregation

method for MIL models. However, considering the complexity of the information within the bag, and the lack of direct supervision over the instances, relying only on the attention mechanism to learn bag-level representations is inadequate. Therefore, we introduce the instance-level classification task to fully use the available information and guide the learning process more effectively. For this purpose, a two-stream feature aggregation module (TSFA) is constructed to enable more optimal integration and utilization of features. The overall pipeline of the TSFA is shown in Fig. 2(b), which jointly learns bag-level embedding representations and key instance selection in a two-stream architecture.

Firstly, we use the gated attention (Ilse et al., 2018) in the TSFA to derive bag-level embedding representations. Given a set of feature vectors $F_{s_p}^{(i)} = \{f_{1s_p}^{(i)}, f_{2s_p}^{(i)}, \dots, f_{ms_p}^{(i)}\}$ at magnification s_p from i -th bag, the attention score for the k -th instance is given by the gated attention (Eq. (1)):

$$\alpha_{ks_p}^{(i)} = \frac{\exp\{w^T \tanh(V f_{ks_p}^{(i)T}) \odot \text{sigm}(U f_{ks_p}^{(i)T})\}}{\sum_{j=1} \exp\{w^T (\tanh(V f_{js_p}^{(i)T})) \odot \text{sigm}(U f_{js_p}^{(i)T})\}}, \quad (1)$$

where w , U and V are all trainable parameters, $\odot$ denotes element-wise multiplication, $\tanh(\cdot)$ is the hyperbolic tangent function, and $\text{sigm}(\cdot)$ is the sigmoid function. In contrast to general attention mechanisms, gated attention introduces learnable non-linearity that overcomes the limitations of linear equations. The embedding representation $d_{s_p}^{(i)}$ of bag $\mathbf{B}_i$ at magnification s_p is obtained from the attention scores, by the following equation:

$$d_{s_p}^{(i)} = \sum_{k=1} \alpha_{ks_p}^{(i)} f_{ks_p}^{(i)}. \quad (2)$$

Concurrent with the obtaining of bag-level embedding representations, an instance-selected branch is designed to select instances for the patch-level classification task and construct pseudo-labels for them. In order to make the distribution of the selected instances as consistent as possible with the distribution of the original patches and ensure that the selected patches are more diversified, a clustering algorithm (e.g. K-means) is first used to divide the input set $F_{s_p}^{(i)}$ into K clusters $S = \{S_{s_p}^{(ik)}\}_{k=1}^K$, i.e.,

$$\{S_{s_p}^{(ik)}\}_{k=1}^K, S_{s_p}^{(ik)} \subseteq F_{s_p}^{(i)} \leftarrow \text{Cluster}(F_{s_p}^{(i)}). \quad (3)$$

Then, the instances in each cluster are sorted according to their distance to the clustering center $c_{ks_p}^{(i)}$ and the nearest instance $\hat{f}_{js_p}^{(ik)}$ is selected as the representative instance $r_{ks_p}^{(i)}$ of that cluster, i.e.,

$$R^{(i)} = \{r_{ks_p}^{(i)} | r_{ks_p}^{(i)} = \hat{f}_{js_p}^{(ik)}, \hat{f}_{js_p}^{(ik)} \in S_{s_p}^{(ik)}\} \leftarrow \text{top}(d(c_{ks_p}^{(i)}, f_{js_p}^{(ik)})), \\ \hat{f}_{js_p}^{(ik)} \in S_{s_p}^{(ik)}, k = 1, 2, \dots, K. \quad (4)$$

where $d(\cdot, \cdot)$ is the Euclidean distance.

For the positive bag, the former $K/2$ instances with the highest attention scores among the selected instances are considered positive instances, while the remaining instances are considered negative ones. For the negative bag, all selected instances are considered negative instances by default. Followed the Top- K instances selection, the patch-level classification is performed through one fully connected layer. A high loss function value of the patch-level classifier indicates that the pseudo-label assignment of patches may be incorrect, i.e., they have been assigned the incorrect attention weights. Therefore, the model is facilitated to learn a more appropriate attention score by the change of loss function value.

In summary, TSFA collaboratively learns bag-level embedding representations and key instances by leveraging the attention weights of instances. Specifically, TSFA employs the weighted summation of attention weights with instances to obtain bag-level representations. Meanwhile, after selecting representative instances using a clustering algorithm, pseudo-labels are assigned to these selected instances based on their attention weights, facilitating the learning of the patch-level classifier.

3.3. Multi-scale contrastive learning

Most current multi-instance contrastive learning methods are patch-level methods based on image enhancement strategies, aiming to extract discriminative information from patches. Moreover, they all require pre-training of the contrastive learning model using patches, which is time-consuming. Therefore, this paper aims to construct a bag-level contrastive learning model to learn the discriminative information of bags and fuse it throughout training, reducing the time required to pre-train the patch-level contrastive learning models. The overall pipeline of the multi-scale contrastive learning (MSCL) is shown in Fig. 2(c).

One key component of contrastive learning is to construct logical positive/negative pairs for training. Different from previous methods of constructing pairs of contrastive learning samples based on image enhancement or selecting the subsets of patches, this paper utilizes the pyramid hierarchical structure of WSI to learn the similarity between bags. The MSCL regards images from the same WSI at different magnifications as positive sample pairs and images from different WSIs as negative samples. In detail, for the i -th bag, the embeddings $d_{s_p}^{(i)}, d_{s_q}^{(i)}$ from its different magnifications constitute positive sample pairs, while with samples $d_{s_k}^{(j)}$ from other WSIs constitute negative sample pairs. These embeddings from different layers of the WSI can be regarded as different views of the same WSI, and thus can be used to form positive pairs in contrastive learning. After obtaining the embedding $d_{s_p}^{(i)}$ of the bag through the TSFA, the projection head maps it onto the space of contrastive learning, obtaining $p_{s_p}^{(i)}$.

To make a difference between the other WSIs and the i th WSI, motivated by Ciga et al. (2022), we apply NT-Xent loss (Chen et al., 2020) as an objective function to maximize the similarity between positive sample pairs and minimize the similarity between negative sample pairs. Then the contrastive learning loss can be calculated as follows:

$$l_{pq}^{(i)} = -\log \frac{\exp(\text{sim}(p_{s_p}^{(i)}, p_{s_q}^{(i)})/\tau)}{\sum_{j \neq i} \sum_{k=1} \exp(\text{sim}(p_{s_p}^{(i)}, p_{s_k}^{(j)})/\tau)}. \quad (5)$$

The τ represents the temperature parameter and the $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity of two vectors. The numerator of the loss function represents the degree of similarity between the sample $p_{s_p}^{(i)}$ and its corresponding positive sample $p_{s_q}^{(i)}$, with the objective of promoting greater similarity between positive samples. In contrast, the denominator represents the degree of similarity between the sample $p_{s_p}^{(i)}$ and all other negative samples $p_{s_k}^{(j)}$, with the objective of minimizing similarity between negative samples.

3.4. Loss function

Overall, the final loss function of proposed method is composed of a contrastive loss L_{CL} and a classification loss L_{CLS} . Here, the contrastive loss L_{CL} can be computed by Eq. (5), and the classification loss L_{CLS} consists of two components, i.e., the bag-level classification loss L_{CLS_B} and the patch-level classification loss L_{CLS_I} .

After the TSFA, we obtain the embedding representation of the bag as well as the filtered instances and their pseudo-labels. They are fed into the bag-level classifier $B(\cdot)$ and instance-level classifier $I(\cdot)$ to get the prediction, respectively. Thus, the classification loss function is as follows:

$$L_{CLS} = \beta L_{CLS_B} + (1 - \beta) L_{CLS_I}, \quad (6)$$

where L_{CLS_B} and L_{CLS_I} are both cross-entropy loss functions, β is a weighting coefficient to balance the bag-level and instance-level classification.

Finally, we can obtain the proposed loss function as follows:

$$L = \alpha L_{CL} + (1 - \alpha) L_{CLS}, \quad (7)$$

where α is a weighting coefficient to balance the contrastive learning and classification. We present the detailed training process of the proposed framework in Algorithm 1.

Algorithm 1: The training process of the proposed MSCL-MIL

Input: Batch size N , Temperature τ , Hyper-parameter α, β, K , Structure of feature extractor $F(\cdot)$, Feature aggregator $\mathcal{A}(\cdot)$, Projection head $\mathcal{P}(\cdot)$, Patch-level classifier $I(\cdot)$ and Bag-level classifier $B(\cdot)$

Output: $\mathcal{A}(\cdot)$, $\mathcal{P}(\cdot)$, $I(\cdot)$, $B(\cdot)$

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1 for batch  $\{\mathbf{B}_n, \mathbf{Y}_n\}_{n=1}^N$  do
2   for all  $n \in \{1, \dots, N\}$  do
3     // Extract the feature vectors of patches at different
       magnifications;
4      $f_{is_p}^{(n)} = F(X_{is_p}^{(n)})$ ,  $f_{js_q}^{(n)} = F(X_{js_q}^{(n)})$ ,
        $i = 1, 2, \dots, m_1$ ;  $j = 1, 2, \dots, m_2$ ;
5     // TSFA;
6     // Compute the attention scores of patches according
       to Eq. (1);
7      $\alpha_{is_p}^{(n)} = \mathcal{A}(f_{is_p}^{(n)})$ ,  $\alpha_{js_q}^{(n)} = \mathcal{A}(f_{js_q}^{(n)})$ ;
8     // Compute the embeddings at different magnifications
       by Eq. (2);
9      $d_{s_p}^{(n)} = \sum_{i=1}^K \alpha_{is_p}^{(n)} f_{is_p}^{(n)}$ ,  $d_{s_q}^{(n)} = \sum_{j=1}^K \alpha_{js_q}^{(n)} f_{js_q}^{(n)}$ ;
10    // Divide the set of feature vectors into  $K$  clusters;
11     $\{S_{s_p}^{(nk)}\}_{k=1}^K, S_{s_q}^{(nk)} \subseteq F_{s_p}^{(n)} \leftarrow \text{Cluster}(F_{s_p}^{(n)}, s_p \geq s_q$ ;
12    // Select patches and construct the pseudo-labels for
       patches;
13     $R^{(i)} = \{r_{ks_p}^{(i)} | r_{ks_p}^{(i)} = \hat{f}_{s_p}^{(ik)}, \hat{f}_{s_p}^{(ik)} \in S_{s_p}^{(ik)}\} \leftarrow$ 
        $\text{top}(d(c_{ks_p}^{(i)}, f_{js_p}^{(ik)}), \hat{f}_{js_p}^{(ik)} \in S_{s_p}^{(ik)}, k = 1, 2, \dots, K$ ;
14    if  $Y_n = 1$  then
15       $y_n = 1$  if  $r_{ks_p}^{(n)} \in \frac{1}{2} \text{top}(R^{(n)})$ , otherwise  $y_n = -1$ ;
16    else
17       $\forall r_{ks_p}^{(n)} \in R^{(n)}, y_n = -1$ ;
18    end
19    // Compute the patch-level classification loss;
20     $L_{CLS_I} \leftarrow \text{Cross-entropy}(I(r_{ks_p}^{(n)}), y_n)$ ;
21    // Compute the bag-level classification loss;
22     $L_{CLS_B} \leftarrow \text{Cross-entropy}(B(d_{s_p}^{(n)}), Y_n)$ 
23  end
24  // MSCL;
25  // Compute the projections of the embeddings at different
       magnifications;
26   $p_{s_p}^{(n)} = \mathcal{P}(d_{s_p}^{(n)})$ ,  $p_{s_q}^{(n)} = \mathcal{P}(d_{s_q}^{(n)})$ ,  $n = 1, \dots, N$ ;
27  // Compute the contrastive loss by Eq. (5);
28   $L_{CL} \leftarrow l_{pq}^{(n)}$ ;
29  Update the  $\mathcal{A}(\cdot)$ ,  $\mathcal{P}(\cdot)$ ,  $I(\cdot)$ ,  $B(\cdot)$  to minimize the loss
       function Eq. (7)
30 end
  
```

4. Experimental setup

This section provides a detailed description of the experimental setup. First, we present the details of the used public dataset as well as the processing of the datasets. Second, we describe the implementation details and evaluation metrics for the proposed method. Finally, we introduce influential methods in the WSIs classification problem to compare with our proposed method in the subsequent sections.

4.1. Datasets

To demonstrate the superior performance of the proposed method, various experiments are conducted over three public histopathology WSI datasets: Camelyon16, The Cancer Genome Atlas (TCGA) non-small cell lung cancer (NSCLC) and The Cancer Genome Atlas (TCGA) esophageal carcinoma (ESCA).

Camelyon16 is a public dataset proposed for metastasis detection in breast cancer. The dataset consists of 270 annotated WSIs for training and another 129 WSIs as a held-out official test set collected at the Radboud University Medical Center and the University Medical Center Utrecht in the Netherlands. After pre-processing, there are a total of about 18 million patches at 40 $\times$ magnification and 1.2 million patches at 10 $\times$ magnification, averaging about 45135 and 2990 patches per bag.

TCGA-NSCLC includes two subtypes of lung cancer, i.e., Lung Squamous Cell Carcinoma (LUSC) and Lung Adenocarcinoma (LUAD), with in a total of 1017 diagnostic digital slides that can be downloaded from National Cancer Institute Data Portal. There are 505 LUAD WSIs from 446 cases and 512 LUSC WSIs from 478 cases. The dataset yields 13 million patches at 20 $\times$ magnification and 0.86 million patches at 5 $\times$ magnification with in average about 13129 and 857 patches per bag.

TCGA-ESCA consists of two types of esophageal carcinoma, for a total of 136 diagnostic WSIs with 73 squamous cell carcinoma from 71 cases and 63 adenocarcinoma from 63 cases, which can be downloaded from National Cancer Institute Data Portal. Following pre-processing, the dataset obtains a total of 7.2 million patches at 40 $\times$ magnification and 0.43 million patches at 10 $\times$ magnification, which averages 47052 patches and 3198 patches per bag.

During the preprocessing of the data, the image at each magnification of the WSIs is first converted from RGB space to HSV space, and foreground segmentation is performed using median filtering with the OTSU algorithm. Then, each WSI is exhaustively cropped non-overlapping patches with the size of 256×256 from within the foreground contours at the specified magnification and stores stacks of image patches along with their coordinates and the slide metadata using the hdf5 hierarchical data format. For all datasets, we randomly split the datasets into training, validation, and testing sets in the ratio of 65 : 15 : 20, respectively.

4.2. Implementations details and evaluation metrics

As in CLAM (Lu et al., 2021b), the feature of each patch is embedded in a 1024-dimensional vector by a ResNet50 (He et al., 2016) pre-trained on ImageNet, followed by a fully connected layer to reduce it from 1024 to 512. The model parameters are updated via the Adam optimizer with a learning rate of 2×10^{-4} and L_2 weight decay of 1×10^{-5} . In addition, the batch size N is set to 12 in our experiments and the temperature τ is set as 0.02. Our models are trained for at least 50 epochs and up to a maximum of 200 epochs. During training, the validation loss in each epoch is observed and the early stopping principle is used when it had not decreased from the previous low point for more than 20 consecutive epochs. In addition, weighted sampling is used in the training process. Specifically, the weights are calculated and assigned based on the number of samples within each category, and subsequent random sampling is conducted based on these weights. The three key parameters, i.e., α in Eq. (7), β in Eq. (6) and the number of clusters K , are then adjusted experimentally, as detailed in Section 5.3.

It is worth noting that our proposed framework can be implemented at multi-scale theoretically, however, we select images at two magnifications of each dataset for our experiments and only use the instance-level branch on the feature aggregator at the highest magnification level due to the limitations of the equipment.

For all the experiments, area under the curve (AUC) is the primary performance metric to report. As the AUC index considers the classification ability of the classifier for both positive and negative samples, it can still objectively evaluate the classifier in the case of imbalanced samples. Besides, the slide-level accuracy (ACC) is also considered. All experiments were conducted on one NVIDIA Tesla V100 GPU with 32 GB Memory.

4.3. Methods for comparison

To better illustrate the advantages of the proposed method, we compared it with the following popular methods, including eight methods based on single magnification and three methods based on multiple magnifications.

Eight related MIL methods based on single magnification for WSIs classification in the literature are considered for our comparative study, termed as MAX-MIL, ABMIL (Ilse et al., 2018), CLAM-SB (Lu et al., 2021b), CLAM-MB (Lu et al., 2021b), DTFD-MIL (MAXS) (Zhang et al., 2022b), DTFD-MIL (AFS) (Zhang et al., 2022b), TransMIL (Shao et al., 2021), and CWC-T (Wang et al., 2023). MAX-MIL is a baseline for MIL, which aggregates patch-level scores by max-pooling to obtain slide-level labels. AB-MIL (Ilse et al., 2018) is the most well-known attention-based MIL method. CLAM (Lu et al., 2021b) introduces attention-based learning and further designs instance-level clustering constrain to refine the feature space, which is divided into CLAM-SB and CLAM-MB based on the number of output layers. Among them, CLAM-SB denotes that there is only one output layer, and CLAM-MB denotes that the number of output layers is the number of classes. DTFD-MIL (Zhang et al., 2022b) virtually expands the number of bags by introducing the concept of pseudo-bags and constructs a two-layer MIL framework based on it. And it is divided into DTFD-MIL (MAXS) and DTFD-MIL (AFS) according to the ways of feature distillation. TransMIL (Shao et al., 2021) introduces the powerful vision transformer architecture to capture the relationship of instances. CWC-T (Wang et al., 2023) combines the convolutional neural network with the Transformer to learn local and global information about WSI.

Three MIL methods based on multiple magnifications are DSMIL (Li et al., 2021), MSCNN (Zhang et al., 2022a) and HAG-MIL (Xiong et al., 2023a). After obtaining the feature vectors of patches at multiple magnifications individually, DSMIL (Li et al., 2021) concatenates and fuses the features through a pyramid strategy, further engaging in the subsequent classification. MSCNN (Zhang et al., 2022a) integrates the attention matrix of patches at multiple magnifications by bilinear transformation, thus realizing multi-scale feature fusion. HAG-MIL (Xiong et al., 2023b) employs the scores derived from the refined transformer model to filter the patches at low magnification, subsequently identifying the corresponding patches at high magnification. Then, the gated attention mechanism is utilized to obtain the slide-level representation.

5. Results

This section provides a comparative assessment of the proposed method for WSI classification. First, we evaluate the classification performance of our proposed MSCL-MIL and compare it with popular methods on three datasets. Second, we conduct ablations to examine the impact of various components in our framework. In addition, we analyze the hyper-parameters used in the model and consider the effect of different training data on the model performance. Finally, we perform visualizations on the Camelyon16 dataset to further demonstrate the results of our method. These results confirm that MSCL-MIL has a more stable performance in predicting WSIs.

5.1. Quantitative comparison

The results of various methods on three datasets are shown in Table 1. The best results are highlighted in bold.

We first compare the performance of Camelyon16 and TCGA-NSCLC datasets, the most popular WSI benchmark classification datasets. As demonstrated in Table 1, the overall prediction ACC/AUC of the MSCL-MIL can achieve 92.5%/95.8% on Camelyon16, and 87.1%/95.5% on TCGA-NSCLC, respectively. The results show that MSCL-MIL exhibits superior performance compared to the other comparative models on Camelyon16, and achieves the best AUC on TCGA-NSCLC. Among them, the improvement of the performance on Camelyon16 is higher

than that on TCGA-NSCLC. This is mainly due to the fact that most of the positive slides on Camelyon16 contain only a small portion of the positive region, thus it is more effective to utilize patch-level classifier to assist the model training.

In addition, it can be observed from Table 1 that MSCL-MIL outperforms most of the methods on TCGA-ESCA. But the AUC of MSCL-MIL is 88.2%, which is slightly lower than that of TransMIL, whereas the ACC value of 82.1% for MSCL-MIL is improved by 7.1% compared to 75% for TransMIL. While the performance of TCGA-ESCA is overall lower than the performance of both TCGA-NSCLC and Camelyon16 datasets. This may be attributed to the limited quantity of slides on TCGA-ESCA, which prevented the models from extracting enough effective information. The experimental results in Section 5.4 further confirm that the amount of training data has an impact on the model performance.

Furthermore, the t-test is employed as a statistical method to ascertain the comparability of the two paired results. The two results are statistically significantly disparate when the p -value is less than 0.05. Table 1 reports the p -values obtained by t-tests for MSCL-MIL compared two-by-two with other comparison methods. The p -value for MSCL-MIL with CLAM-MB and DTFD-MIL is less than 0.05, indicating that MSCL-MIL is statistically significantly different from the aforementioned two methods.

Overall, from the results of the above comparison experiments, it can be seen that our method exhibits competitive performance for the problem of classifying WSIs of different cancers, indicating that our method has better generalization.

5.2. Ablation experiments

We conducted ablation studies to demonstrate the effectiveness of our proposed MSCL module and an instance-level classifier based on the TSFA module. The main results of the ablation experiments are given in Table 2.

Effect of TSFA. To evaluate the effectiveness of the TSFA components, we conducted the following experiments on three datasets. M_{Att} indicates the model that employs a bag-level classifier with only a gated attention-based mechanism. M_{CLS_I} implies the model that incorporates the patch-level classifier based on the above bag-level classifier, where $M_{CLS_{I1}}$ denotes the training of the patch-level classifier by filtering the instances only by attention scores, while $M_{CLS_{I2}}$ trains the patch-level classifier using the TSFA module of this paper. From the Table 2, it can first be observed that M_{CLS_I} outperforms M_{Att} in terms of performance at the same magnification. This observation indicates that incorporating the patch-level classifier to augment the supervisory characteristics of the model can potentially enhance its overall performance. Furthermore, regarding the TSFA module is concerned, it can be seen from the table that the TSFA in $M_{CLS_{I2}}$ contributes more to the performance of the model than filtering the instances using only the attention scores in $M_{CLS_{I1}}$. This improvement may be attributed to the clustering algorithm, which brings the distribution of filtered instances closer to that of the original data and exposes the model to different patches.

Effect of MSCL. M_{MSCL} denotes the model that is trained using our proposed contrastive learning strategy along with the bag-level classifier, where the positive/negative pairs are from images at different magnifications of the slides. As shown in the Table 2, M_{MSCL} achieves better results than M_{Att} , demonstrating the importance of considering discriminative information between bags. In Table 2, MSCL-MIL achieves the best performance than all ablation models in most cases, demonstrating the capability of the whole framework we propose.

In addition, we explore the effect of within-scale contrastive learning loss, named $M_{SSCL-MIL}$. We sample subsets of patches from the bags to construct positive/negative sample pairs for contrastive learning at a single scale. Positive sample pairs are constructed by randomly sampling one-quarter of the total patches from one slide at a single scale, following the methodology described in the literature (Tavolara

Table 1

The performance of different MIL approaches on Camelyon16, TCGA-NSCLC and TCGA-ESCA datasets. The highest performance is in bold.

Method	Scale	Camelyon16		TCGA-NSCLC		TCGA-ESCA		<i>p</i> -value
		ACC	AUC	ACC	AUC	ACC	AUC	
MAX-MIL	Single	0.900	0.915	0.841	0.899	0.750	0.744	0.117
ABMIL	Single	0.863	0.815	0.851	0.914	0.786	0.836	0.147
CLAM-SB	Single	0.813	0.893	0.876	0.940	0.812	0.790	0.126
CLAM-MB	Single	0.888	0.945	0.861	0.934	0.821	0.867	0.021
TransMIL	Single	0.913	0.939	0.871	0.936	0.750	0.892	0.436
DTFD-MIL(MAXS)	Single	0.887	0.921	0.851	0.924	0.786	0.841	0.006
DTFD-MIL(AFS)	Single	0.887	0.930	0.886	0.950	0.821	0.862	0.119
CWC-T ^a	Single	0.891	0.939	0.859	0.949	–	–	0.305
DSMIL ^a	Multiple	0.856	0.899	0.888	0.939	–	–	0.331
HAG-MIL ^a	Multiple	0.887	0.946	0.849	0.921	–	–	0.284
MSCNN ^a	Multiple	0.923	0.934	–	–	–	–	–
MSCL-MIL (ours)	Multiple	0.925	0.958	0.871	0.955	0.821	0.882	–

^a Results are taken from the papers (Zhang et al., 2022b,a; Xiong et al., 2023b; Wang et al., 2023).**Table 2**

Ablation results for evaluating the TSFA and MSCL in MSC-MIL on three datasets. Research on TSFA ablation involves using attention mechanisms for bag embeddings (Attention), selecting key instances for patch-level classification via attention scores (CLS_I), and a proposed clustering-based instance selection method for the same purpose (Cluster). The ablation study on MSCL is about proposed contrastive learning loss (CL).

Camelyon16							
Method	Scale	Attention	CLS_I	Cluster	CL	ACC	AUC
M_{At}	40×	✓				0.888	0.912
$M_{CLS_{I1}}$	40×	✓	✓			0.875	0.918
$M_{CLS_{I2}}$	40×	✓	✓	✓		0.913	0.923
M_{MSCL}	10×, 40×	✓			✓	0.913	0.943
$M_{SSCL-MIL}$	40×	✓	✓	✓	✓	0.900	0.919
MSCL-MIL	10×, 40×	✓	✓	✓	✓	0.925	0.958
TCGA-NSCLC							
Method	Scale	Attention	CLS_I	Cluster	CL	ACC	AUC
M_{At}	20×	✓				0.841	0.931
$M_{CLS_{I1}}$	20×	✓	✓			0.847	0.933
$M_{CLS_{I2}}$	20×	✓	✓	✓		0.866	0.936
M_{MSCL}	5×, 20×	✓			✓	0.861	0.950
$M_{SSCL-MIL}$	20×	✓	✓	✓	✓	0.866	0.942
MSCL-MIL	5×, 20×	✓	✓	✓	✓	0.871	0.955
TCGA-ESCA							
Method	Scale	Attention	CLS_I	Cluster	CL	ACC	AUC
M_{At}	40×	✓				0.786	0.764
$M_{CLS_{I1}}$	40×	✓	✓			0.786	0.805
$M_{CLS_{I2}}$	40×	✓	✓	✓		0.821	0.795
M_{MSCL}	10×, 40×	✓			✓	0.750	0.836
$M_{SSCL-MIL}$	40×	✓	✓	✓	✓	0.821	0.872
MSCL-MIL	10×, 40×	✓	✓	✓	✓	0.821	0.882

et al., 2022). Negative sample pairs are composed of subsets from different slides. The results in Table 2 show that MSCL-MIL outperforms $M_{SSCL-MIL}$, which may be because our constructed positive/negative sample pairs at multiple scales are better able to learn the discriminative information.

Effect of Contrastive Learning Loss. We compare the NT-Xent-based contrastive learning loss function proposed in the paper with other commonly used contrastive learning loss functions on Camelyon16, including NT-Logistic loss (Mikolov et al., 2013) and Margin Triple loss (Schroff et al., 2015). Table 3 shows that the NT-Xent-based loss we use gives the best results, which may be due to the fact that cosine similarity and temperature can effectively weight different samples, and appropriate temperature can help the model learn from hard negatives (Chen et al., 2020).

5.3. Hyper-parameters study

There are three key hyper-parameters, i.e., α in Eq. (7), β in Eq. (6) and the number of clusters K , that need to be adjusted during the

Table 3

Evaluating results of MSCL with different contrastive loss functions on Camelyon16.

	Margin triple	NT-Logistic	NT-Xent
AUC	0.938	0.936	0.958

training process of our proposed method. We analyze the choice of hyper-parameters on three datasets.

During the training process, we vary $\alpha \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, $\beta \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$ and $K \in \{6, 8, 10, 12, 14\}$ to train the model, respectively, where we fix two parameters to tune the other, rather than considering the combination of all hyper-parameters. The results are illustrated in Fig. 3, where (a), (b) and (c) illustrate the effect of the weight coefficients α , β and the number of clusters K on the model on the three datasets, respectively. As can be seen from Fig. 3(a) and (c), for the Camelyon16 and TCGA-NSCLC datasets, α and K do not have a significant effect on the model, i.e., the AUCs of the proposed method are all around 0.95 on the datasets with different values, which are better than most of the methods in Table 1. In the case of the TCGA-ESCA dataset, the two hyper-parameters, α and K , have a greater impact on the model, and the optimal values of the hyper-parameters are $\alpha = 0.5$ and $K = 12$, respectively. In addition, the weight of the contrastive loss has a more noticeable effect on the model performance improvement on the TCGA-ESCA dataset compared to the other two datasets. This difference is most likely because the fact that the TCGA-ESCA dataset contains a smaller amount of data, resulting in the need for the model to increase the weight of the contrastive learning loss to capture the discriminative features of the WSIs more effectively. As for the hyper-parameter β , it can be observed from Fig. 3(b) that the optimal value for all three datasets is 0.8.

5.4. Effect of data ratios

The requirement of using image features extracted at different magnifications in our proposed model means that the input information obtained by the whole model is doubled. To explore whether our model can achieve comparable performance that other methods with single magnification when even the ratio of the training set is reduced, we conduct experiments using the Camelyon16 dataset to validate our idea, and the results are shown in Fig. 4.

When the ratio of the training set is reduced to 75%, our model achieves competitive results compared to some models based on single magnification. Even when the ratio of the training set is reduced to 50%, a very comparable result can still be obtained, i.e., the AUC is only reduced by 0.6% compared to using the full training data. However, when the ratio of our training set drops further below 50% (about 129 WSIs), the performance of the model decreases rapidly with the reduced amount of data. This means that our model has a good ability to learn discriminative information about pathological slides,

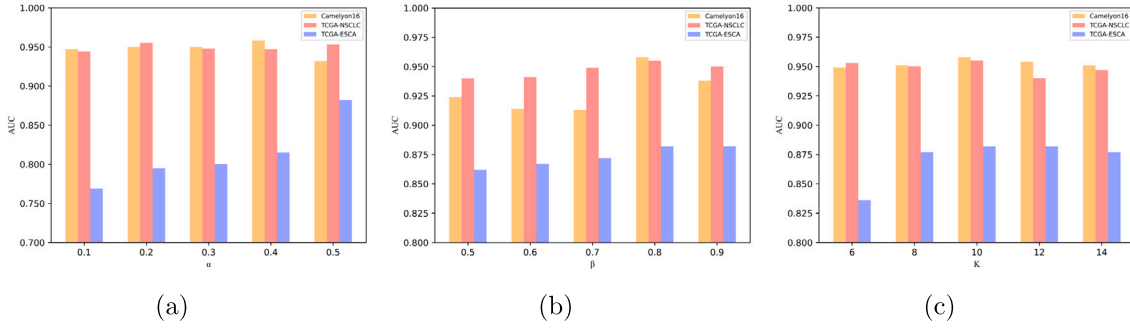


Fig. 3. Results on three datasets for different hyper-parameters. The x-axis represents the corresponding parameter, and the y-axis represents the AUC value.

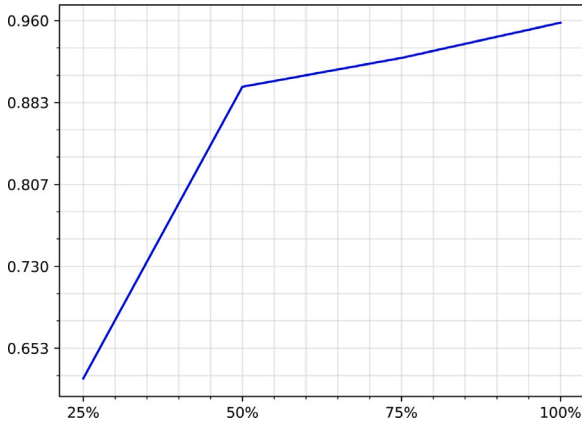


Fig. 4. Results on Camelyon16 dataset of MSCL-MIL using 100, 75, 50 and 25% of cases in the training set. The x-axis represents the training dataset ratio, and the y-axis represents the AUC value.

making it robust enough to maintain competitive performance even with appropriately reduced datasets, but if the amount of data is too small, the performance suffers.

5.5. Visualization and interpretation

Some visualization results on the Camelyon16 dataset are presented in this subsection in order to provide an intuitive interpretation of our method. Fig. 5 shows the visualization results of the slide-level features learned by several well-performing methods on the testing set. Due to the high and varying feature dimensions of the various methods, the t-SNE algorithm (Van der Maaten and Hinton, 2008) is used to reduce the feature dimensions to 2D for visualization. In the figure, the red color indicates the tumor slides, and the purple color indicates normal slides. As observed from the figure, the features learned by our proposed method are more separable.

In addition, we visualize the contribution of each patch in a WSI to the classification decision through an attention heatmap. As shown in Fig. 6(a), the area in the green curve is the tumour region, which is visualized according to the annotations provided by the Camelyon16 dataset. In Fig. 6(c), the attention scores of our proposed method are visualized as a heatmap to identify the regions of interest, where interpret regions of high attention are displayed in red (high contribution to the prediction of the WSI relative to other patches) and low attention displayed in blue (low contribution to the prediction of the WSI relative to other patches). Fig. 6(b) is an enlarged view of the color squares in Fig. 6(a) and Fig. 6(c). It is observed that there is a high consistency between the labeled region and the obtained heatmap, illustrating the good interpretability of the proposed method.

6. Discussion

In this work, we develop a novel multi-scale multi-instance contrastive learning approach for the problem of classifying whole slide images. The key technological innovations are the design of a two-stream feature aggregator and a novel contrastive learning training paradigm based on bag-level embedding representations. Compared to current state-of-the-art MIL methods, our model exhibits competitive performance on the Camelyon16, TCGA-NSCLC, and TCGA-ESCA datasets.

The pyramid structure inherent to WSI contains information at varying levels of magnification, with the lower magnification capturing coarse-grained information and the higher magnification capturing fine-grained cellular details. Pathologists often need to combine observations at multiple magnifications to make a final diagnosis. It is therefore plausible that the excellent performance of the MSCL-MIL model may be attributed to the fact that its consistency with pathologist behavior. First, we utilize the inherent pyramid structure of WSI to treat images originating from the same slide but with different magnifications as positive sample pairs, and images from different slides as negative sample pairs. The implementation of a multi-scale contrastive learning loss strategy encourages the model to learn not only the multi-view consistency features exhibited by images with different magnifications of the same slide, but also to discriminate between the discriminative features of different slide images during the learning process. The results of the ablation experiments in Table 2 further confirm that training with multi-scale features has a positive effect on improving the overall performance of the model. Besides, the introduction of the patch-level classifier enhances the supervisory of the model, enabling it to better learn and evaluate the importance of patches. As shown in Table 2, the patch-level classifier proposed in this paper exhibits enhanced performance in both the AUC and ACC metrics compared to the model that only uses a bag-level classifier.

While MSCL-MIL has shown commendable classification results in a variety of WSI datasets, limitations remain. Specifically, there is the issue of computational cost associated with the use of information at multiple magnifications. Table 4 presents the training time and inference time of MSCL-MIL with some comparison methods on the Camelyon16 and TCGA-NSCLC datasets. It can be observed that although the inference time for trained MSCL-MIL is comparable to other methods, the total training time required by MSCL-MIL is about 10 to 50 times longer than other methods, where the training time per epoch also reaches 15 to 30 times longer than other methods. This phenomenon is mainly due to the high computational load imposed by the use of information at multiple magnifications in MSCL-MIL. Therefore, the design of a lightweight multi-instance learning model that can effectively utilize the hierarchical information of WSIs remains an open problem and a promising area for future research.

With regard to practical applications, our work has the potential to be utilized in clinical settings where annotation limitations exist.

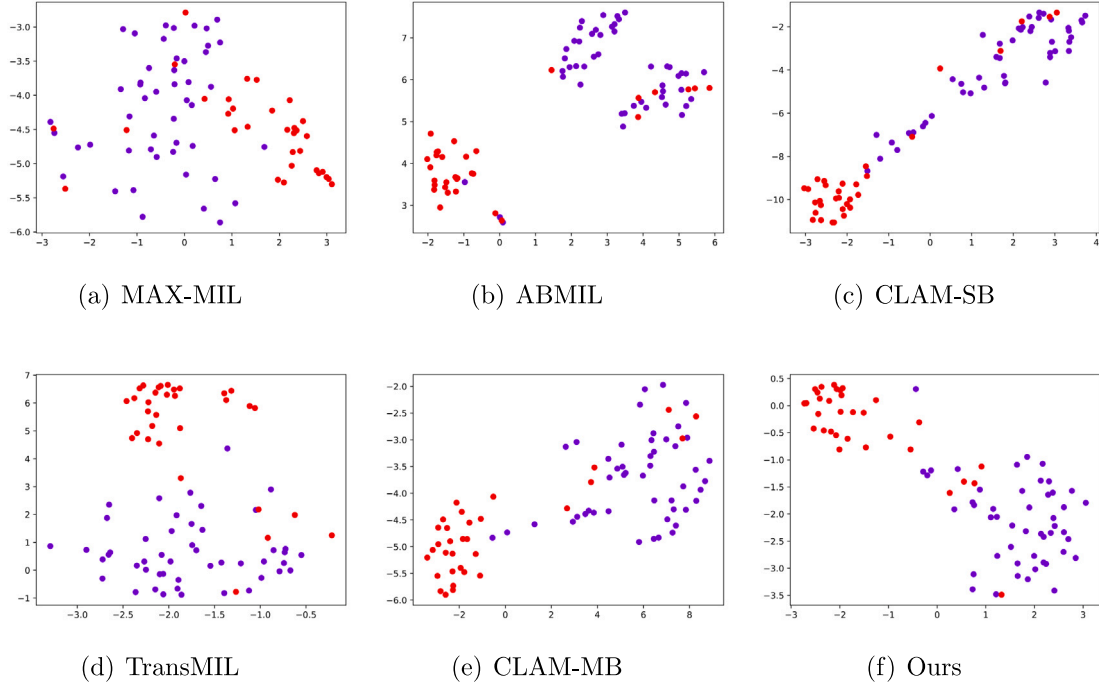


Fig. 5. Visualization of slide-level features learned by the various models using t-SNE on the testing set of Camelyon16.

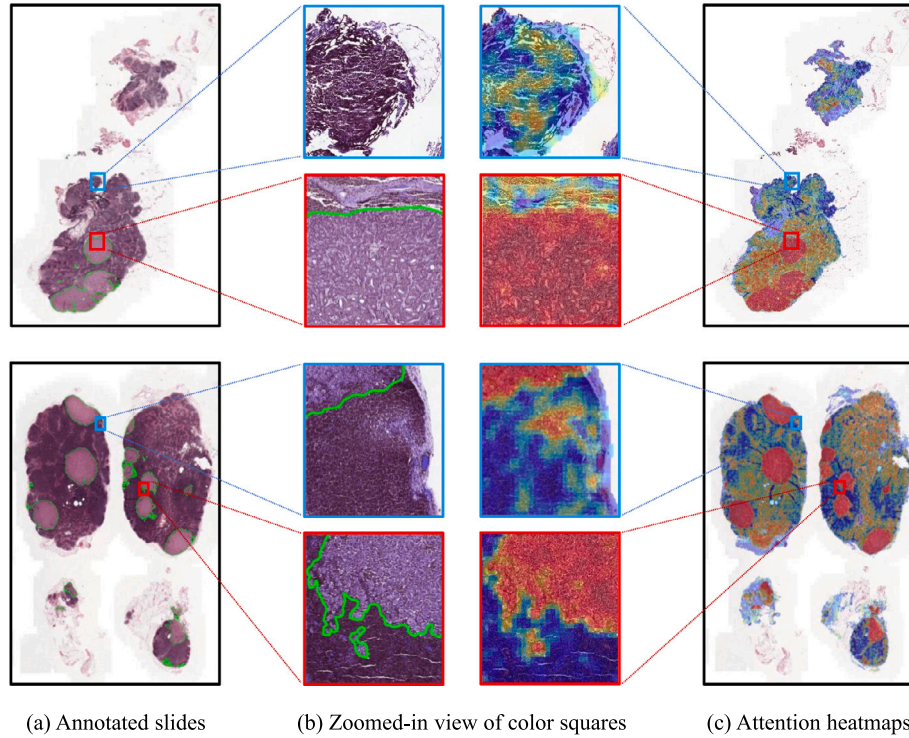


Fig. 6. Examples of attention maps obtained from the Camelyon16 dataset. (a) Tumor annotation provided by raw data. The green curve represents the localization annotation. (b) Enlarged view of the colored squares in (a) and (c). (c) Attention maps.

The process of WSI analysis is tedious, laborious, and time-consuming due to the large size of the WSIs. Deploying a trained model in a clinical setting can facilitate more expedient and accurate diagnoses, reducing the burden on medical professionals. Specifically, our trained model can automate the analysis of a multitude of pathology slide images, quickly identify potential tumor cases, and present intuitive visual analysis results to assist doctors in making judgments, improving the efficiency and precision of tumor screening.

7. Conclusion

In this paper, we propose a novel MIL model based on multi-scale contrastive learning for the classification of WSIs. Comparison experiments are conducted on three publicly available datasets to validate the effectiveness of the method. In addition, ablation experiments, hyper-parametric analysis, discussion on the effect of data ratios, and visualization experiments are performed to analyze our method in

Table 4
Comparison of training time and inference time for different MIL methods on Camelyon16 and TCGA-NSCLC datasets. The total training time (Total) , training time per epoch (Train), and inference time (Inference) are reported.

Method	Camelyon16			TCGA-NSCLC		
	Total (s)	Train (s)	Inference (s)	Total (s)	Train (s)	Inference (s)
MAX-MIL	1367.9711	26.3071	7.3011	1029.8329	19.8045	5.6279
ABMIL	1366.9173	26.2869	7.2982	1178.5752	22.6649	5.7343
CLAM-SB	1344.9546	25.8645	6.1009	1356.7285	26.0909	5.8987
CLAM-MB	1303.0781	24.5864	6.1545	1399.4152	26.9118	6.0591
TransMIL	2115.1743	88.1323	13.2211	1240.3173	72.9598	13.4676
DTFD-MIL(MAXS)	6467.1435	32.3357	10.8672	6274.4947	31.3725	6.3297
DTFD-MIL(AFS)	6502.0469	32.5102	9.9626	5855.4453	29.2772	5.8494
MSCL-MIL (ours)	65 698.8597	413.2004	10.3721	55 664.0547	639.8167	8.5692

depth. Although the proposed MSCL-MIL has obtained promising performance on three WSI datasets, it still has some limitations. Specifically, the utilization of multi-scale information significantly increases computational and storage requirements. Furthermore, the proposed model was originally designed for classification tasks without considering other tasks such as survival analysis. In future work, we would consider developing lightweight multi-instance learning models to reduce hardware requirements. Additionally, we will further explore the application of the proposed approach to survival analysis.

CRedit authorship contribution statement

Jianan Zhang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Fang Hao:** Writing – review & editing, Visualization, Formal analysis. **Xueyu Liu:** Writing – review & editing, Formal analysis. **Shupeí Yao:** Writing – review & editing. **Yongfei Wu:** Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis. **Míng Lì:** Supervision, Formal analysis. **Wen Zheng:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used are publicly available datasets.

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